# Solar C appendix

Sun · C · downloadable appendix

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|---|---|
| **Document** | Solar C appendix |
| **Subject** | Solar photospheric abundance appendix · C |
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Solar carbon: 8.536 ± 0.042

Asplund, Amarsi & Grevesse 2021: 8.46 ± 0.04 · this measurement +0.076 dex

Combined the way Asplund, Amarsi & Grevesse (2021) combine it: weighted mean and error of the indicator families [C I], C I, C2, CH, CO (Asplund+2021 Sect. 2.3); 1D results moved to the 3D scale by Asplund+2021 Table 3 (3D minus 1D MARCS for the same indicator). Each indicator is represented by our best analysis of it.

C I: 8.488 ± 0.049 · IAG FTS · red-optical · C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)

C2: 8.441 ± 0.078 (1D 8.419 +0.022 to 3D) · HARPS · VIS · $C_2$ Swan band (molecular band) · 1D LTE

CH: 8.575 ± 0.034 (1D 8.509 +0.066 to 3D) · HARPS · VIS · CH G-band (molecular band) · 1D LTE

CO: 8.559 ± 0.126 (1D 8.701 -0.142 to 3D) · CRIRES+ · K · MOL-CO_K · 1D LTE

Not measurable in our spectra: [C I].

## Highlighted Products

### VIS

**8.463 ± 0.039**

C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · HARPS · Reference Grade

### red-optical

**8.488 ± 0.049**

C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · IAG FTS · Reference Grade

### NIR

**8.559 ± 0.036**

C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

### H

**8.396 ± 0.120**

SET-ELGUETA26 · 1D LTE · CRIRES+ H (full arm, telluric-corrected) · Reference Grade

### K

**8.701 ± 0.126**

MOL-CO_K · 1D LTE · CRIRES+ · Codex Grade

## Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

| Grade | What it means | This element |
|---|---|---|
| Reference Grade | lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used | 29 products -- C I · Asplund et al. 2021 lines; SET-ELGUETA26 |
| Codex Grade | lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep | 31 products -- C I 5052 Å (single line); C I 5380 Å (single line); C I · Codex-graded lines; CH G-band (molecular band); $C_2$ Swan band (molecular band); MOL-CO_K |
| Deep Grade | the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere | none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines) |

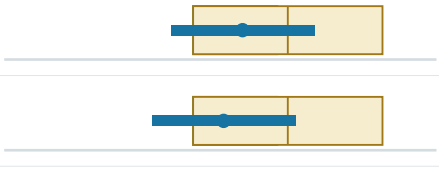
## Error-bar Forest Plot

### VIS — HARPS
### A(C) dex; axis 8.24–8.65



| | |
|---|---|
| C I 5052 Å (single line) · 1D LTE<br>Codex Grade · 1 line | 8.444<br>±0.095 |
| C I 5380 Å (single line) · 1D LTE<br>Codex Grade · 1 line | 8.528<br>±0.093 |
| $C_2$ Swan band (molecular band) · 1D LTE<br>Codex Grade · 167 lines | 8.419<br>±0.078 |
| CH G-band (molecular band) · 1D LTE<br>Codex Grade · 184 lines | 8.509<br>±0.034 |
| C I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 3 lines | 8.483<br>±0.039 |
| C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 3 lines | 8.463<br>±0.039 |
| C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 3 lines | 8.476<br>±0.039 |
| C I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 2 lines | 8.476<br>±0.046 |
| C I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 2 lines | 8.488<br>±0.047 |

### VIS — Kitt Peak (Kurucz 2005)
### A(C) dex; axis 8.24–8.65



| | |
|---|---|
| C I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 3 lines | 8.467<br>±0.068 |
| C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 3 lines | 8.449<br>±0.068 |

## VIS — Kitt Peak (Kurucz 2005)
## A(C) dex; axis 8.24–8.65

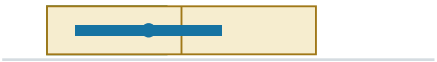
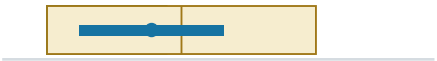
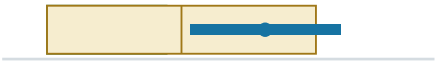
| | | |
|---|---|---|
| C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 3 lines |  | 8.460<br>±0.068 |
| C I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 2 lines |  | 8.363<br>±0.093 |
| C I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 2 lines |  | 8.375<br>±0.091 |

## VIS — Kitt Peak (1984, molecfit)
## A(C) dex; axis 8.24–8.65

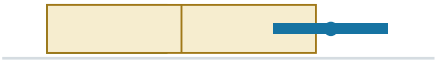
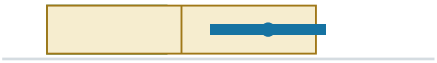
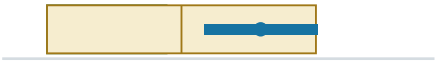
| | | |
|---|---|---|
| C I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 3 lines |  | 8.450<br>±0.062 |
| C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 3 lines |  | 8.430<br>±0.062 |
| C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 3 lines |  | 8.443<br>±0.062 |
| C I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 2 lines |  | 8.376<br>±0.092 |
| C I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 2 lines |  | 8.388<br>±0.090 |

VIS axis: 8.24 · 8.34 · 8.45 · 8.55 · 8.65 A(C) dex

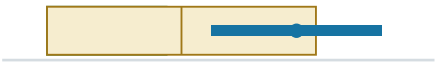
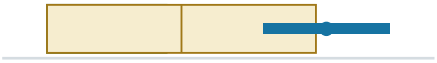
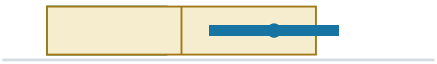
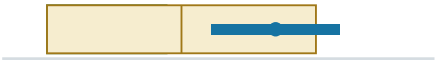
## red-optical — IAG FTS
## A(C) dex; axis 8.39–8.68

| | | |
|---|---|---|
| C I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 10 lines |  | 8.529<br>±0.051 |
| C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 10 lines |  | 8.488<br>±0.049 |
| C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 10 lines |  | 8.490<br>±0.048 |
| C I · Codex-graded lines · 1D LTE<br>Codex Grade · 6 lines |  | 8.566<br>±0.050 |
| C I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 6 lines |  | 8.528<br>±0.052 |
| C I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 6 lines |  | 8.527<br>±0.051 |

## red-optical — Kitt Peak (Kurucz 2005)
## A(C) dex; axis 8.39–8.68

| | | |
|---|---|---|
| C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 10 lines |  | 8.536<br>±0.053 |
| C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 10 lines |  | 8.532<br>±0.053 |
| C I · Codex-graded lines · 1D LTE<br>Codex Grade · 6 lines |  | 8.610<br>±0.038 |
| C I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 6 lines |  | 8.568<br>±0.039 |
| C I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 6 lines |  | 8.563<br>±0.038 |

## red-optical — Kitt Peak (1984, molecfit)
## A(C) dex; axis 8.39–8.68

| | | |
|---|---|---|
| C I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 10 lines |  | 8.587<br>±0.057 |
| C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 10 lines |  | 8.541<br>±0.055 |
| C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 10 lines |  | 8.540<br>±0.053 |
| C I · Codex-graded lines · 1D LTE<br>Codex Grade · 6 lines |  | 8.607<br>±0.042 |
| C I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 6 lines |  | 8.572<br>±0.043 |
| C I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 6 lines |  | 8.573<br>±0.043 |

red-optical axis: 8.39 · 8.46 · 8.53 · 8.61 · 8.68 A(C) dex

## NIR — CRIRES+ J
## A(C) dex; axis 8.39–8.78

| | | |
|---|---|---|
| SET-ELGUETA26 · 1D LTE<br>Reference Grade · 12 lines |  | 8.488<br>±0.040 |
| C I · Codex-graded lines · 1D LTE<br>Codex Grade · 2 lines |  | 8.521<br>±0.038 |

## NIR — CRIRES+ Y (full arm)
## A(C) dex; axis 8.39–8.78

| | | |
|---|---|---|
| SET-ELGUETA26 · 1D LTE<br>Reference Grade · 6 lines |  | 8.569<br>±0.035 |

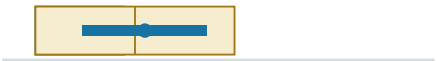
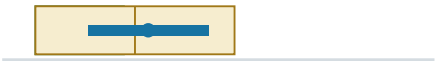
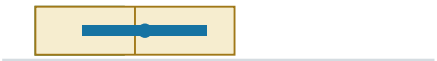
## NIR — CRIRES+ Y (full arm)
A(C) dex; axis 8.39–8.78

| | | |
|---|---|---|
| C I · Codex-graded lines · 1D LTE<br>Codex Grade · 4 lines |  | 8.580<br>±0.037 |

## NIR — IAG FTS
A(C) dex; axis 8.39–8.78

| | | |
|---|---|---|
| C I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 4 lines |  | 8.677<br>±0.074 |
| C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 3 lines |  | 8.602<br>±0.089 |
| C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 3 lines |  | 8.600<br>±0.086 |
| C I · Codex-graded lines · 1D LTE<br>Codex Grade · 4 lines |  | 8.677<br>±0.074 |
| C I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 3 lines |  | 8.602<br>±0.089 |
| C I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 3 lines |  | 8.600<br>±0.086 |

## NIR — Kitt Peak (Kurucz 2005)
A(C) dex; axis 8.39–8.78

| | | |
|---|---|---|
| C I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 4 lines |  | 8.607<br>±0.057 |
| C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 3 lines |  | 8.519<br>±0.056 |
| C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 3 lines |  | 8.522<br>±0.055 |
| C I · Codex-graded lines · 1D LTE<br>Codex Grade · 4 lines |  | 8.607<br>±0.057 |
| C I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 3 lines |  | 8.519<br>±0.056 |
| C I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 3 lines |  | 8.522<br>±0.055 |

## NIR — Kitt Peak (1984, molecfit)
A(C) dex; axis 8.39–8.78

| | | |
|---|---|---|
| C I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 4 lines |  | 8.623<br>±0.035 |
| C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 3 lines |  | 8.559<br>±0.036 |

### NIR — Kitt Peak (1984, molecfit)
### A(C) dex; axis 8.39–8.78

| | | |
|---|---|---|
| C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 3 lines |  | 8.561<br>±0.035 |
| C I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 3 lines |  | 8.559<br>±0.036 |
| C I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 3 lines |  | 8.561<br>±0.035 |

NIR axis: 8.39 · 8.49 · 8.59 · 8.68 · 8.78 A(C) dex

### H — CRIRES+ H (full arm, telluric-corrected)
### A(C) dex; axis 8.18–8.63

| | | |
|---|---|---|
| SET-ELGUETA26 · 1D LTE<br>Reference Grade · 4 lines |  | 8.396<br>±0.120 |
| C I · Codex-graded lines · 1D LTE<br>Codex Grade · 3 lines |  | 8.367<br>±0.161 |

H axis: 8.18 · 8.29 · 8.40 · 8.52 · 8.63 A(C) dex

### K — CRIRES+
### A(C) dex; axis 8.39–8.86

| | | |
|---|---|---|
| MOL-CO_K · 1D LTE<br>Codex Grade · 64 lines |  | 8.701<br>±0.126 |

K axis: 8.39 · 8.51 · 8.62 · 8.74 · 8.86 A(C) dex

**How to read this forest.** Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Each bar is the uncertainty as Asplund et al. 2021 quote it: line-to-line scatter, the 3D-model term and the NLTE term (at least 0.03 dex). For a molecule the NLTE term is replaced by its dependence on the other element, and the 3D term comes from the 3D-to-1D difference Amarsi et al. 2021 publish for the same band. Atomic-data and observational terms are not in the bar, as in Asplund; they stay in the full error budget attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

## How this was measured

Every carbon value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the carbon abundance as the free parameter. The synthesis includes the atomic lines and the molecular bands (CN, CH, $C_2$, OH and others) in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, blends and coupling to the other C/N/O abundances, and the spread between spectra, each measured

by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

| | |
|---|---|
| Band | VIS |
| What was measured | C I · Codex-graded lines |
| Solar spectra | HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-NLTE, 3D-NLTE |
| Lines | 2 |
| Result A(C) | 8.363–8.476<br>(3D-NLTE) |
| Largest uncertainty terms (dex) | continuum placement 0.05, line strengths (gf) 0.04 |

| | |
|---|---|
| Band | VIS |
| What was measured | C I 5052 Å (single line) |
| Solar spectra | HARPS |
| Treatments | 1D-LTE |
| Lines | 1 |
| Result A(C) | 8.444<br>(1D-LTE) |
| Largest uncertainty terms (dex) | continuum placement 0.09, fit / line-to-line scatter 0.09 |

| | |
|---|---|
| Band | VIS |
| What was measured | C I 5380 Å (single line) |
| Solar spectra | HARPS |
| Treatments | 1D-LTE |
| Lines | 1 |
| Result A(C) | 8.528<br>(1D-LTE) |
| Largest uncertainty terms (dex) | continuum placement 0.15, spread between spectra 0.12 |

| | |
|---|---|
| Band | VIS |
| What was measured | $C_2$ Swan band (molecular band) |
| Solar spectra | HARPS |
| Treatments | 1D-LTE |
| Lines | 167 |
| Result A(C) | 8.419<br>(1D-LTE) |
| Largest uncertainty terms (dex) | line strengths (gf) 0.13, microturbulence 0.12 |

| | |
|---|---|
| Band | VIS |
| What was measured | CH G-band (molecular band) |
| Solar spectra | HARPS |
| Treatments | 1D-LTE |
| Lines | 184 |
| Result A(C) | 8.509<br>(1D-LTE) |

| | |
|---|---|
| Band | VIS |
| What was measured | CH G-band (molecular band) |
| Largest uncertainty terms (dex) | line strengths (gf) 0.11, spread between spectra 0.06 |

| | |
|---|---|
| Band | VIS |
| What was measured | C I · Asplund et al. 2021 lines |
| Solar spectra | HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, 3D-NLTE |
| Lines | 3 |
| Result A(C) | 8.430–8.463<br>(3D-NLTE) |
| Largest uncertainty terms (dex) | continuum placement 0.10, fit window 0.03 |

| | |
|---|---|
| Band | red-optical |
| What was measured | C I · Codex-graded lines |
| Solar spectra | IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, 3D-NLTE |
| Lines | 6 |
| Result A(C) | 8.528–8.572<br>(3D-NLTE) |
| Largest uncertainty terms (dex) | telluric residual 0.09, fit window 0.06 |

| | |
|---|---|
| Band | red-optical |
| What was measured | C I · Asplund et al. 2021 lines |
| Solar spectra | IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, 3D-NLTE |
| Lines | 10 |
| Result A(C) | 8.488–8.541<br>(3D-NLTE) |
| Largest uncertainty terms (dex) | fit / line-to-line scatter 0.04, NLTE / 3D correction 0.03 |

| | |
|---|---|
| Band | NIR |
| What was measured | C I · Codex-graded lines |
| Solar spectra | CRIRES+ J, CRIRES+ Y (full arm), IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, 3D-NLTE |
| Lines | 2–4 |
| Result A(C) | 8.519–8.602<br>(3D-NLTE) |
| Largest uncertainty terms (dex) | telluric residual 0.04, fit window 0.04 |

| | |
|---|---|
| Band | NIR |
| What was measured | C I · Asplund et al. 2021 lines |
| Solar spectra | IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) |
| Treatments | 1D-LTE, 1D-NLTE, 3D-NLTE |
| Lines | 3–4 |

| | |
|---|---|
| Band | NIR |
| What was measured | C I · Asplund et al. 2021 lines |
| Result A(C) | 8.519–8.602<br>(3D-NLTE) |
| Largest uncertainty terms (dex) | telluric residual 0.04, fit window 0.04 |

| | |
|---|---|
| Band | NIR |
| What was measured | SET-ELGUETA26 |
| Solar spectra | CRIRES+ J, CRIRES+ Y (full arm) |
| Treatments | 1D-LTE |
| Lines | 6–12 |
| Result A(C) | 8.488–8.569<br>(1D-LTE) |
| Largest uncertainty terms (dex) | telluric residual 0.05, fit window 0.05 |

| | |
|---|---|
| Band | H |
| What was measured | C I · Codex-graded lines |
| Solar spectra | CRIRES+ H (full arm, telluric-corrected) |
| Treatments | 1D-LTE |
| Lines | 3 |
| Result A(C) | 8.367<br>(1D-LTE) |
| Largest uncertainty terms (dex) | telluric residual 0.44, fit window 0.21 |

| | |
|---|---|
| Band | H |
| What was measured | SET-ELGUETA26 |
| Solar spectra | CRIRES+ H (full arm, telluric-corrected) |
| Treatments | 1D-LTE |
| Lines | 4 |
| Result A(C) | 8.396<br>(1D-LTE) |
| Largest uncertainty terms (dex) | fit / line-to-line scatter 0.11, telluric residual 0.11 |

| | |
|---|---|
| Band | K |
| What was measured | MOL-CO_K |
| Solar spectra | CRIRES+ |
| Treatments | 1D-LTE |
| Lines | 64 |
| Result A(C) | 8.701<br>(1D-LTE) |
| Largest uncertainty terms (dex) | model atmosphere 0.14, coupling to the other C/N/O abundances 0.08 |

## Excluded lines

These lines are left out of the Sun's measurement. They stay in the line lists: a cooler or more evolved star may measure them well. Blended lines are not excluded -- the spectrum synthesis models the blend. 1 lines.

**Saturated**: on the flat part of the curve of growth, so the line barely responds to the abundance (reduced width log W/λ above -4.9).

| Species | Wavelength (Å, air) | Why it is excluded |
|---|---|---|
| C I | 5380.340 | saturated |

## Solar carbon context

Carbon is traced by atomic C I, including forbidden [C I], and molecular CH, $C_2$ and CO indicators. Each probes different line formation and molecular-equilibrium sensitivities. Indicator families and atmospheric treatments remain separate products.

C/O indicator strategy · Methodology

## Download the evidence

Download PDF — Solar C appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

## Reproducibility

Generated from the pinned science snapshot. Source commit
e76c06c6173928306b433a805509e63a1a206c71
Feed C.json v1.203 · updated 2026-10-29T03:00:31Z
Generated 2026-11-05T21:50:45-07:00 (source commit timestamp)
Feed SHA-256 8532ab51e95fbd5a737f9f61f0028f6531d0d9396ee155d20e25a22130947a3a
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

## References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

**Solar abundance and indicator analyses.** Solar C atomic/molecular indicator lineage and atmospheric sensitivity; comparison literature, not an adopted Codex abundance. Sources: [1], [2].

**Atomic data and gf authorities.** The recorded C I/II pedigree is the partial NIST MCHF update described by Wiese & Fuhr; evaluated theory is not primary laboratory evidence. Sources: [3], [4].

**Atomic data and gf authorities.** AGSS21 names Li et al. for its C I gf lineage. This reference calculation is distinct from the campaign's NIST/MCHF adjudication; no gf source is substituted here. Sources: [5].

**Model and grid sources.** Registered C I line/grid lineage: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — line-by-line 3D-NLTE / 1D-NLTE grids (CDS J/A+A/630/A104). The current feed treatment remains authoritative; a grid reference does not imply a correction was applied. Sources: [6].

**Atomic data and gf authorities.** Atomic C I entry; source-family pedigree only, not a substitute for a per-transition gf join. Sources: [3], [4].

**Model and grid sources.** Feed treatment: 1D-LTE; no NLTE/3D correction is claimed for this entry. Turbospectrum is the synthesis implementation. Sources: [7].

**Molecular line data and delivery.** CO line-data delivery: CO ExoMol line list, Li et al. 2015 (ApJS 216, 15). RYA-236 conversion of the ExoMol Li2015 CO list to the Turbospectrum babsma .dat format (converted to species code 0608.012016; conversion script external to this repo — RYA-236) ExoMol/Turbospectrum is a redistribution/conversion, not the primary publication. Sources: [2], [7], [8].

**Model and grid sources.** Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 3D non-LTE - 1D LTE on STAGGER (table2 C I / table3 O I); PER-LINE additive correction, grid lines only (match <= 0.05 A) Sources: [6].

**Model and grid sources.** Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 1D non-LTE - 1D LTE (table5 C I / table6 O I); PER-LINE additive correction, grid lines only (match <= 0.05 A) Sources: [6].

**Molecular line data and delivery.** C2 line-data delivery: C2 Swan $d^3\Pi g–a^3\Pi u$ (Brooke et al. 2013 / Masseron compilation). Masseron compilation, VALD-based, distributed with iSpec/Turbospectrum (Gerber et al. 2023, A&A 669, A43; uploaded 2022 to keeper.mpdl.mpg.de, split per wavelength region) — see molecules-dir README.md Sources: [2], [7], [9].

**Molecular line data and delivery.** CH line-data delivery: CH A-X: Masseron et al. 2014 (MNRAS 442, 1489) / 2022 update. Masseron compilation, VALD-based, distributed with iSpec/Turbospectrum (Gerber et al. 2023, A&A 669, A43; uploaded 2022 to keeper.mpdl.mpg.de, split per wavelength region) — see molecules-dir README.md The primary intake identifies Masseron 2014 as A&A 571 A47; the older delivery manifest's MNRAS locator conflicts with that record. Sources: [2], [7], [10].

Product-to-source index

H · CRIRES+ H (full arm, telluric-corrected) · SET-ELGUETA26 · 1D LTE · [1], [3], [4], [7] C-products-f04678f420c3

H · CRIRES+ H (full arm, telluric-corrected) · C I · Codex-graded lines · 1D LTE · [1], [3], [4], [7] C-products-7015fc1e2bcc

K · CRIRES+ · MOL-CO_K · 1D LTE · [1], [2], [7], [8] C-products-a3286d05d53c

NIR · CRIRES+ J · SET-ELGUETA26 · 1D LTE · [1], [3], [4], [7] C-products-9b721d3f7ec5

NIR · CRIRES+ J · C I · Codex-graded lines · 1D LTE · [1], [3], [4], [7] C-products-ed9b75aee66f

NIR · CRIRES+ Y (full arm) · SET-ELGUETA26 · 1D LTE · [1], [3], [4], [7] C-products-6449cb5086f8

NIR · CRIRES+ Y (full arm) · C I · Codex-graded lines · 1D LTE · [1], [3], [4], [7] C-products-52eb6ae4e11b

NIR · IAG FTS · C I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [7] C-products-65dcaa1cf605

NIR · IAG FTS · C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-b80d1c35e7d3

NIR · IAG FTS · C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-fe537374f04a

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NIR · IAG FTS · C I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-f2e0374fd8bc

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NIR · Kitt Peak (Kurucz 2005) · C I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-8585afef7622

NIR · Kitt Peak (1984, molecfit) · C I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [7] C-products-d60eacfa6f91

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NIR · Kitt Peak (1984, molecfit) · C I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-d66637bbc168

VIS · HARPS · C I 5052 Å (single line) · 1D LTE · [1], [3], [4], [7] C-products-7fc96af7bda6

VIS · HARPS · C I 5380 Å (single line) · 1D LTE · [1], [3], [4], [7] C-products-536db2a5fb17

VIS · HARPS · $C_2$ Swan band (molecular band) · 1D LTE · [1], [2], [7], [9] C-products-4309ba5d0384

VIS · HARPS · CH G-band (molecular band) · 1D LTE · [1], [2], [7], [10] C-products-19d90d71f47b

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VIS · Kitt Peak (1984, molecfit) · C I · Codex-graded lines · 1D LTE · [1], [3], [4], [7] C-quarantine-9b3b9069e4e7

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**[4]** — Wiese, W. L.; Fuhr, J. R. (2006). New Critical Compilations of Atomic Transition Probabilities for Neutral and Singly Ionized Carbon, Nitrogen, and Iron. Proceedings of the NASA Laboratory Astrophysics Workshop (14-16 February 2006, UNLV, Las Vegas) source record

**[5]** — Li, Amarsi, Papoulia, Ekman & Jonsson 2021, MNRAS 502, 3780 DOI: 10.1093/mnras/stab214. source record

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**[7]** — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. A&A 669, A43 DOI: 10.1051/0004-6361/202243673. source record

### Molecular line data and delivery

**[8]** — Li et al. 2015, ApJS 216, 15 DOI: 10.1088/0067-0049/216/1/15. source record

**[9]** — Brooke et al. 2013, JQSRT 124, 11 DOI: 10.1016/j.jqsrt.2013.02.025. source record

**[10]** — Masseron et al. 2014, A&A 571 A47 DOI: 10.1051/0004-6361/201423956. source record